Functional Annotation Report: *lpd* (PP_5366, UniProt Q88C17) in *Pseudomonas putida* KT2440

Target: Dihydrolipoyl dehydrogenase (dihydrolipoamide dehydrogenase, E3), EC 1.8.1.4 **Gene:** *lpd* / OrderedLocusName **PP_5366** (GenBank AAN70931.1; RefSeq WP_010955849.1) **Organism:** *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) **Length:** 466 aa, ~49.4 kDa; genome position 6,115,792–6,117,192

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1. Summary (Answer to the Research Question)

The product of **PP_5366 (*lpd*, Q88C17)** is a **dihydrolipoyl dehydrogenase (dihydrolipoamide dehydrogenase, "E3", EC 1.8.1.4)** — an FAD-dependent, homodimeric flavoenzyme of the class-I pyridine-nucleotide–disulfide oxidoreductase family. Its primary catalytic function is to **reoxidize the reduced (dihydro)lipoyl group carried on the lysine of a partner E2/H-protein and transfer the electrons, via a redox-active active-site disulfide and the FAD, to NAD^+, producing NADH** (N^6-[(R)-dihydrolipoyl]-L-lysyl-[protein] + NAD^+ → N^6-[(R)-lipoyl]-L-lysyl-[protein] + NADH + H^+). It operates in the **cytoplasm** as a soluble homodimer.

A key identification result: although the KT2440 genome labels PP_5366 generically as "*lpd*", **sequence and genomic-context analysis show it is specifically the ortholog of the historically described "third lipoamide dehydrogenase," LPD-3 (gene *lpd3*)** — 97.0% identical to *P. putida* LPD-3, but only ~51% identical to the operon-encoded E3 of the pyruvate/2-oxoglutarate dehydrogenases (LpdG) and ~45% to that of the branched-chain keto-acid dehydrogenase (LpdV). Those two complex-dedicated E3s are separate genes in KT2440 (**PP_4187 lpdG** and **PP_4404 lpdV**). PP_5366/LPD-3 is a standalone, eukaryote-like isozyme that is catalytically competent (it can fully substitute for the housekeeping E3 in pyruvate dehydrogenase and ~60% in 2-oxoglutarate dehydrogenase) but whose dedicated in-vivo physiological niche remains uncharacterized.

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2. Gene/Protein Identity Verification (Mandatory)

- **Gene symbol match:** ✔ "lpd" is the accepted symbol for lipoamide/dihydrolipoyl dehydrogenase; the protein description (Dihydrolipoyl dehydrogenase, EC 1.8.1.4) is consistent.
- **Organism match:** ✔ *P. putida* KT2440 (NCBI taxid 160488).
- **Family/domain match:** ✔ Class-I pyridine nucleotide–disulfide oxidoreductase; FAD/NAD-binding + dimerization domains; lipoamide-DH signature (InterPro IPR006258, IPR050151, IPR036188, IPR023753, IPR016156; NCBIfam TIGR01350 lipoamide_DH; PROSITE PS00076).
- **Paralog disambiguation (critical):** ✔ Resolved. KT2440 encodes **three** EC 1.8.1.4 paralogs (KEGG KO K00382): PP_4187 (*lpdG*), PP_4404 (*lpdV*), and PP_5366 (*lpd* = *lpd3* ortholog). Pairwise identity of PP_5366: **97.0%** to LPD-3 (P31046), 51.1% to LpdG (P31052), 45.2% to LpdV (P09063). PP_5366's 1401-bp/466-aa coding region exactly matches the reported *lpd3* [PMID 1722146].

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3. Molecular Function — What Reaction Is Catalyzed and Substrate Specificity

Reaction (UniProt/Rhea:15045, EC 1.8.1.4): N^6-[(R)-dihydrolipoyl]-L-lysyl-[protein] + NAD^+ ⇌ N^6-[(R)-lipoyl]-L-lysyl-[protein] + NADH + H^+

- **Substrate specificity:** the physiological substrate is the **protein-bound dihydrolipoyl group** (the lipoyl-lysine "swinging arm" of an E2 dihydrolipoyl acyltransferase or of the glycine-cleavage H-protein), with **NAD^+ as the electron acceptor**. The isolated LPD-3 protein was shown biochemically to be "clearly a lipoamide dehydrogenase as opposed to a mercuric reductase or glutathione reductase" [PMID 2914869], confirming the specificity within this mechanistically related flavoprotein family.
- **Cofactor:** one non-covalently bound **FAD per subunit** (UniProt; ChEBI:57692).
- **Active site / mechanism:** a **redox-active disulfide (Cys42–Cys47)** accepts electrons from dihydrolipoamide and passes them to FAD, which then reduces NAD^+. Structural studies of the conserved family show the mechanism "uses two molecules: non-covalently bound FAD and a transiently bound substrate, NAD^+," and that in the reductive half-reaction "the nicotinamide base stacks directly on the isoalloxazine ring system of the FAD" for hydride transfer [PMID 15946682]. A conserved **His (His445 here) proton acceptor**, working with a

neighboring Glu, deprotonates the dithiol during catalysis. Sequence features in Q88C17 confirm this architecture: FAD-binding residues (51, 115, 144–146, 313, 319–322), NAD^+-binding residues (181–188, 204, 273), catalytic His445, and the redox-active Cys42/Cys47 disulfide.

- **Regulation:** family E3s are subject to product inhibition by NADH (over-reduction of FAD); assembly with the E2 core relieves this by lowering flavin redox potential — "subunit interaction plays an important role in the protection of the enzyme against over-reduction" and interaction with E2p/E2o "relieves the inhibition to a large extent" [PMID 8575446].

Functional competence of LPD-3: in a *lpdG* mutant, LPD-3 "completely restored pyruvate dehydrogenase activity" and was "about 60% as effective as LPD-Glc in restoring 2-ketoglutarate dehydrogenase activity" [PMID 2914869], demonstrating it is a fully active E3.

3a. Structure-Based Inference (AlphaFold)

No experimental structure exists for this specific protein, but the AlphaFold model **AF-Q88C17-F1 (v6)** is of exceptional confidence (**global mean pLDDT** = **97.4**; 98% of residues pLDDT ≥ 90; 0% below 50), and every functionally assigned residue is modeled at very high confidence (Cys42 = 98.0, Cys47 = 97.4, FAD-binding Ser51/His115/313 ≈ 98–99, NAD-binding 181/204/273 ≈ 96–98, catalytic His445 = 96.9). Sequence/structure analysis confirms the diagnostic features of the glutathione-reductase-like class-I pyridine-nucleotide–disulfide oxidoreductase fold: - **FAD Rossmann fingerprint** G9-G10-G11-P-G13-G14 (GxGxxG); - **NAD^+ Rossmann fingerprint** G181-A-G183-V-I-G186 (GxGxxG); - **Redox-active disulfide Cys42/Cys47** within the lipoamide-DH signature (…GGT**C**LNVG**C**MPSK…); - **Catalytic His445–Glu450 dyad** (…TC**H**PHPT**RSE**…), the general base of the E3 mechanism; - **Two-domain topology** (N-terminal FAD/NAD(P)-binding Rossmann fold + C-terminal dimerization domain; Pfam PF07992/PF02852).

This independent structural evidence corroborates the enzymatic annotation and places the redox-active dithiol at the *re*-face of the flavin, adjacent to the transient NAD(H) site — the geometry required for the FAD-mediated electron relay [PMID 15946682].

4. Quaternary Structure and Subcellular Localization

- **Quaternary structure: homodimer**; each of the two active sites is built at the dimer interface, and the redox-active disulfide of one subunit is juxtaposed with the FAD (UniProt

SUBUNIT; consistent with all characterized E3 crystal structures, e.g., human E3 [PMID 15946682]).

- **Localization: cytoplasm** (UniProt SUBCELLULAR LOCATION; GO:0005737). As a bacterial soluble enzyme it performs its reaction in the cytosol, physically/dynamically associated with the peripheral E2 cores of the 2-oxo-acid dehydrogenase complexes (a "transferase complex", GO:1990234) or with the glycine-cleavage machinery.

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5. Pathway Context and Biological Process

Dihydrolipoyl dehydrogenase is the shared terminal component of the **lipoyl-dependent oxidative decarboxylation systems**: "LADH is the common E3 subunit of the alpha-ketoglutarate (KGDHc), pyruvate (PDHc), and branched-chain α-keto acid dehydrogenase complexes and is also part of the glycine cleavage system" [PMID 28579060]. By reoxidizing the E2/H-protein lipoyl arm and generating NADH, the enzyme:

- Couples **pyruvate → acetyl-CoA** (PDH; KEGG M00307) and the **2-oxoglutarate → succinyl-CoA** step of the TCA cycle (KGDH; M00009/M00011) to the NAD^+/NADH pool and hence to respiration (GO:0045333 cellular respiration).
- Supports **branched-chain amino-acid catabolism** (BCKAD; leucine degradation M00036) and the **glycine cleavage system** (M00621; one-carbon/folate metabolism).

Division of labor in *P. putida* (important distinction): these physiological roles are carried out by the **operon-encoded, complex-dedicated E3s**, not primarily by PP_5366: - **LpdG (PP_4187)** — encoded within the *sucABCD* operon (neighbors *sucB*/E2o, *sucA*/E1o); it is "the E3 component of the pyruvate and 2-ketoglutarate dehydrogenase complexes and the L-factor for the glycine oxidation system" [PMID 1902462]. - **LpdV (PP_4404)** — encoded within the *bkdAA–bkdAB–bkdB* branched-chain keto-acid dehydrogenase operon; it is "the specific E3 component of the branched-chain keto acid dehydrogenase complex... induced by growth on leucine, isoleucine, or valine" [PMID 1902462], under BkdR/L-branched-chain-amino-acid control [PMID 10217783]. - **PDH (aceEF)** — the pyruvate dehydrogenase E1/E2 genes are located separately at PP_0339 (*aceE*/E1p) and PP_0338 (*aceF*/E2p), far from all three E3 loci; the PDH complex therefore has **no dedicated co-operonic E3** and recruits a shared E3 (LpdG), exactly as the biochemistry indicates. (KEGG lists only three K00382 E3 paralogs genome-wide — PP_4187, PP_4404, PP_5366 — so no fourth, PDH-specific E3 exists.) - **PP_5366 / LPD-3 (this gene)** — the "third lipoamide dehydrogenase... whose role is unknown" [PMID 1902462]; *lpd3* "is not part of an operon, which is unique for a prokaryotic lipoamide dehydrogenase" [PMID 1722146], and the protein "was not produced in wild-type *P. putida*... under a variety of growth conditions" [PMID 1722146]. It is the most eukaryote-like of the three (~54% identity to pig/human mitochondrial E3) [PMID 1722146].

Evolutionary placement (this work): a cross-species pairwise-identity comparison shows PP_5366/LPD-3 is atypically eukaryote-like: 50.9% identical to human E3 (P09622) and 50.7% to yeast E3 (P09624) — as close as to its own bacterial paralog LpdG (51.1%) — whereas the canonical *E. coli* housekeeping Lpd (P0A9P0) is only 43.9% identical to human E3 and 42.9% to LPD-3. This indicates LPD-3 is an evolutionarily distinct isozyme (ancient duplication or horizontal acquisition), consistent with its standalone genomic location, rather than a recent duplicate of LpdG/LpdV.

Interpretation of PP_5366's specific role: its molecular function is unambiguous (an authentic, catalytically competent dihydrolipoamide:NAD^+ oxidoreductase). However, its dedicated physiological niche/regulon is not established. The evidence (latent expression in the wild type, standalone locus, ability to substitute for LpdG) is most consistent with a **backup/redundant isozyme** that can supply E3 activity to the 2-oxo-acid dehydrogenase complexes under conditions where the primary E3 is absent or limiting. The KEGG assignment of PP_5366 to all E3-dependent pathways reflects its enzymatic capability (K00382) rather than a demonstrated dedicated in-vivo assignment.

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6. Supported and Refuted Hypotheses

Hypothesis	Verdict	Basis
PP_5366 encodes a dihydrolipoyl dehydrogenase (EC 1.8.1.4), FAD homodimer, cytoplasmic	**Supported**	UniProt Q88C17; family/domain signatures; biochemistry of LPD-3 [PMID 2914869]
PP_5366 is the operon-embedded, complex-dedicated housekeeping E3 (LpdG)	**Refuted**	Only 51% identity to LpdG; the dedicated E3s are separate genes PP_4187/PP_4404; PP_5366 is standalone
PP_5366 is the branched-chain (LpdV) E3	**Refuted**	Only 45% identity to LpdV; LpdV = PP_4404 in the *bkd* operon
PP_5366 is the ortholog of the "third" lipoamide dehydrogenase, LPD-3/*lpd3*	**Supported**	97.0% identity to LPD-3; exact 466-aa/1401-bp match; standalone locus [PMID 1722146]
PP_5366/LPD-3 is catalytically able to serve 2-oxo-acid dehydrogenase complexes	**Supported**	Complements *lpdG* mutant: full PDH, ~60% OGDH restoration [PMID 2914869]
PP_5366/LPD-3 has a defined, dedicated physiological role	**Not established**	Latent in wild type; role historically "unknown" [PMID 1902462; P 1722146]

7. Evidence Types

- **Experimental (primary):** enzymatic characterization and mutant complementation of LPD-3 [PMID 2914869]; gene cloning/sequence and operon analysis of *lpd3*, *lpdG*, *lpdV* [PMID 1722146, 1902462, 2917566, 6185468]; family catalytic-mechanism crystallography [PMID 15946682]; E3–E2 interaction/inhibition [PMID 8575446]; *bkd* regulation [PMID 10217783].
- **Bioinformatic/sequence (this work):** Needleman–Wunsch pairwise identity assigning PP_5366 to LPD-3 (97.0%); KEGG K00382 paralog enumeration (PP_4187/4404/5366); genomic-context (operon) analysis placing LpdG in *sucABCD* and LpdV in *bkd*, PP_5366

standalone; UniProt/InterPro domain and active-site residue mapping; Rossmann fingerprint and His445–Glu450 dyad identification.

- **Structural inference (this work):** AlphaFold AF-Q88C17-F1 v6, mean pLDDT 97.4, all catalytic/cofactor residues pLDDT 96–99, confirming the canonical E3 fold and active-site architecture.
- **Database annotation:** UniProt Q88C17, KEGG ppu:PP_5366, InterPro, Rhea.

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8. Limitations and Future Directions

- The precise physiological role and expression conditions of the LPD-3 isozyme (PP_5366) remain undetermined; classic work was performed in strain PpG2, not KT2440. Targeted transcriptomics/proteomics and a clean ΔPP_5366 (alone and combined with Δ*lpdG*/Δ*lpdV*) knockout in KT2440 would test whether it provides conditional backup E3 activity, stress-specific function, or a distinct partner.
- No experimental 3D structure of PP_5366 exists; the mechanism/active-site description is inferred from the highly conserved E3 family plus a very-high-confidence AlphaFold model (mean pLDDT 97.4). Because E3 function in KT2440 is distributed across three paralogs (PP_4187/PP_4404/PP_5366), a single-gene knockout is not expected to be phenotypically essential on rich or minimal medium (cf. the genome-wide conditional-essentiality screen of KT2440 [PMID 20158506]); dedicated multi-knockout studies would be needed to expose the specific contribution of PP_5366.
- The eukaryote-like character of LPD-3 raises (untested) questions about horizontal acquisition or a specialized redox/moonlighting function; this was not resolved here.

References (PMID)

1722146 · 1902462 · 2914869 · 2917566 · 6185468 · 15946682 · 17960497 · 8575446 · 28579060 · 10217783 · 20158506
